



Banks and Gesso, LLC

22 April 2003

Erica S. Crosby
Environmental Protection Specialist
Division of Minerals and Geology
Department of Natural Resources
1313 Sherman St., Room 215
Denver, CO 80203

720 Kipling St., Suite 117
Lakewood, Colorado 80215

(303) 274-4277
Fax (303) 274-8329
www.banksandgesso.com

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APR 23 2003

Division of Minerals and Geology

RE: CEMEX, Inc Lyons Mine Permit No. M-1977-208 (SI-01)
Reclamation Cost for Demolition of the Cement Plant

Dear Ms. Crosby,

This letter is in response to your comments dated March 6, 2003 regarding an initial bond calculation for CEMEX, Inc. Lyons Mine Permit No. M-1977-208. A copy of the March 6, 2003 letter is included in Appendix A for your reference. An additional letter with bearing on Reclamation Bond calculation for the site, received from the Division of Minerals and Geology (referred to as Division herein), has also been included in Appendix A. Banks and Gesso, LLC has addressed each of your comments, provided information regarding cost assumption, some of which was included in the previous submittal, and provided a revised demolition and disposal cost estimate.

Response to Comments

Response #1

The office buildings, primarily located in the administration area, are constructed of concrete, and as such, are already included in the cost estimate. See Appendix B for additional concrete volume information. Machinery and building contents do not need to be included in the cost estimate, as stated in the March 6, 2003 letter from the Division. Steel structures, tanks, utility lines and conveyors have been quantified and the cost to demolish and dispose of these items/structures is included in the cost estimate. Calculations are included in Appendix C.

Response #2

No aggregate processing or sale will be assumed.

Response #3

All of the concrete will be disposed of in an onsite disposal area. The area has been delineated on the site plan included herein. Additional information regarding the disposal area has been provided herein. A Technical Revision to the permit will be submitted to the Division within 60 days of posting of the financial warranty as allowed in the April 7, 2003 letter to Cemex from the Division. Steel structures, tanks, utility lines and conveyors will be removed to an off site landfill. The selected landfill is located

approximately 30 miles from the plant site. These disposal costs are included in the cost estimate.

Response #4

Unit costs have been adjusted to Division-provided values where available.

Response #5

An approximate scale has been added to the schematic plan provided for this submittal. However, dimensions were taken from the engineering drawings, not scaled from a plan view map.

Response #6

This column called "concrete in place" on Tables 4 and 5 is the sum of the three adjoining columns: Silos, Buildings, and Foundations. These were itemized because different demolition costs were applied to the three categories. In the following submittal, the "concrete in place" column has not been shown on Tables 4 and 5.

Response #7

As required by federal regulations, CEMEX appropriately disposes hazardous waste within 90 days of when the waste is produced; therefore, no hazardous waste will be on site at the time of plant closure. Chemicals used in blasting and processing will be consumed prior to plant closure.

CEMEX has provided an itemization of hazardous materials currently stored on the site. All of the chemicals are in quantities that are less than the required reportable value by federal regulation. This list is located in Appendix D.

Response #8

Line items including mobilization/demobilization, insurance, administration, engineering, performance bond, and profit have been added in the revised cost estimate.

Cost Estimate Details

The overall cost estimate is provided in Table 1. This includes the subtotaled costs from Tables 2, 3 and 7. Tables 4, 5 and 6 include additional data in support of Table 3.

The site was divided into two areas for the estimate; the north part of the site is referred to as the Grinding portion, and the south part is referred to as the Burning portion. This division between the two processes allows a separate bond calculation in the event that the plant is modified so that only the Grinding process occurs in the future. The drawing included with this report depicts the division.

Included in the grinding portion of the plant are the administration building, the mill building, the central control building, raw material silos, and support buildings. The complete list is included herein on Table 5. Included in the burning portion of the plant are the burner building, the coal unloading building and coal silo, the primary and secondary crushers, the bag houses, and support buildings. The complete list is included herein on Table 6.

Steel

The quantity of the steel structures including conveyors, tanks/hoppers, building, fencing, stairs, grating, and utility lines on site has been estimated. These items are included in the cost estimate as Table 2. The line item costs presented in Table 2 include demolition, transportation and disposal to an offsite disposal location.

Concrete

The concrete demolition and disposal costs estimates are included in Table 3. The calculations for the concrete include all the structures that are above ground surface. Buried foundations, footings, and tunnels were not included in the calculation. However, the building floors and concrete pavement in the plant area were included in the calculation and a uniform thickness of one foot was assumed. The line item costs presented in Table 3 include demolition, transportation and disposal to the onsite disposal area.

The demolition costs for the buildings on the site were calculated using values based on the volume of the buildings as they stand (Table 4). It was assumed that the interior floors, columns, foundations, etc., would require additional work, and thus were assigned a separate demolition value based on the volume of concrete. The line item costs presented in Table 3 include demolition, transportation and disposal to the onsite disposal area.

Disposal

Onsite Disposal

The selected disposal site is along a hogback ridge at the south end of the plant site (see site plan map). The disposal site will accommodate approximately 175,000 cubic yards of concrete. The disposal area will require only minor regrading to receive a geosynthetic liner with the appropriate permeability requirements. The disposal site will be capped with three feet of engineered fill, compacted per specifications and revegetated. The borrow source for the fill is located on the Cemex, Inc. Lyons Mine site, two miles or less from the location of the selected disposal site. Additional information on the disposal site including dimensions, the relationship to groundwater and surface water, and material placement techniques will be addressed in the Technical Revision.

Offsite Disposal Location

The selected disposal site is the RPS Landfill on Highway 93 in Golden. This landfill is located approximately 30 miles south of the Lyons Plant.

Unit Cost Sources

The Division provided some unit costs. These include the unit cost for demolition and disposal of buildings, conveyors, fencing, and utility lines, as well as the unit cost for a

Burnett and Sons, Inc. Concrete Crushing and Recycling in Athens, Texas, Porta-Crushers in Littleton, Colorado, and Alpine Demolition in Arvada, Colorado. Also, Means Heavy Construction Cost Data (2001 Edition) was consulted for some of the items.

Cost Summary

The total cost for demolition and disposal of the entire plant is approximately \$7.67 million. The individual values are presented in Table 1.

Included with this submittal are the tables that summarize the quantities and the costs used for the work. Included as an appendix to this report is a set of more detailed calculation sheets that reference the drawings from which the dimensions of structures were taken. A plan view of the site indicating the process area delineation and labels of structures is also attached.

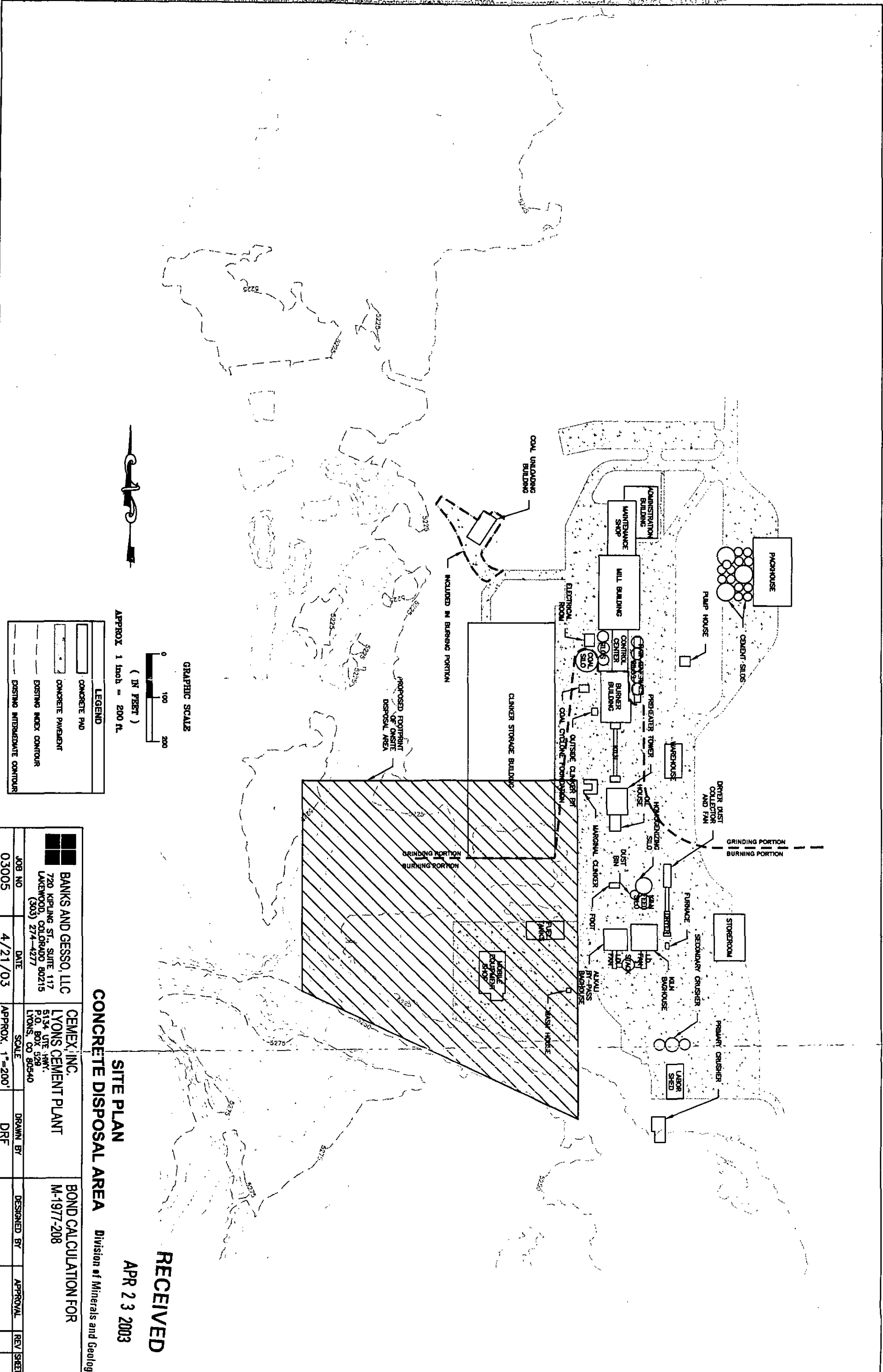
Conclusion

We trust that this submittal provides the Division with the necessary information to process the bond calculation. Please contact us should there be questions regarding this submittal.

Regards,
Banks and Gesso, LLC

A handwritten signature in cursive script, reading "M. Danelle Volpe".

M. Danelle Volpe, P.E.



Cost Tables

**Table 1
Cost Estimate**

CEMEX, Inc Lyons Mine
Permit No. M-1977-208
Bond Calculation

Grinding Process

	Cost
<u>Direct Costs</u>	
On-Site Costs	
Concrete Demolition and Transport to Disposal Site	\$ 2,249,694
Steel Demolition and Transport to Disposal Site	\$ 612,003
Construction of Disposal Site	\$ 56,500
Revegetation	\$ 150,000
Total On-Site Costs	\$ 3,068,197
Mobilization/Demobilization (10% of On-Site Cost)	\$ 306,820
<u>Total Direct Cost</u>	\$ 3,375,017
<u>Indirect Costs</u>	
Liability Insurance (1.55% of Direct Cost)	\$ 52,313
Performance Bond (1.05 of Direct Cost)	\$ 35,438
Profit (10% of Direct Cost)	\$ 337,502
Engineering Work (5% of Direct Cost)	\$ 168,751
Total Indirect Cost	\$ 594,003
<u>Total Cost - Grinding Portion</u>	3,970,000

Burning Process

	Cost
<u>Direct Costs</u>	
On-Site Costs	
Concrete Demolition and Transport to Disposal Site	\$ 1,830,951
Steel Demolition and Transport to Disposal Site	\$ 819,512
Construction of Disposal Site	\$ 56,500
Revegetation	\$ 150,000
Total On-Site Costs	\$ 2,856,963
Mobilization/Demobilization (10% of On-Site Cost)	\$ 285,696
<u>Total Direct Cost</u>	\$ 3,142,660
<u>Indirect Costs</u>	
Liability Insurance (1.55% of Direct Cost)	\$ 48,711
Performance Bond (1.05 of Direct Cost)	\$ 32,998
Profit (10% of Direct Cost)	\$ 314,266
Engineering Work (5% of Direct Cost)	\$ 157,133
Total Indirect Cost	\$ 553,108
<u>Total Cost - Burning Portion</u>	3,700,000

Table 2
Steel Demolition and Reclamation Cost

CEMEX, Inc Lyons Mine
Permit No. M-1977-208
Bond Calculation

Structure	Quantity	Grinding Process		Total Cost
		Units	Cost	
Tanks/Hoppers	508,344	ft ³	\$0.77	\$ 391,425
Utility Lines	2,680	lf	\$26.82	\$ 71,878
Conveyors	800	lf	\$31.85	\$ 25,480
Conveyors Disposal	445	yd ³	\$10.00	\$ 4,448
Metal Buildings	131,000	ft ³	\$0.77	\$ 100,608
Stairs	18,300	ft ³	\$0.77	\$ 14,054
Grating	19,600	ft ²	\$0.18	\$ 3,528
Fencing	300	lf	\$1.94	\$ 582
Total Steel Demolition and Transport Cost - Grinding Portion				\$ 612,003

Structure	Quantity	Burning Process		Total Cost
		Units	Cost	
Tanks/Hoppers	85,357	ft ³	\$0.77	\$ 65,725
Utility Lines	2,680	lf	\$26.82	\$ 71,878
Conveyors	2,700	lf	\$31.85	\$ 85,995
Conveyors Disposal	1,501	yd ³	\$10.00	\$ 15,012
Metal Buildings	748,000	ft ³	\$0.77	\$ 574,464
Stairs	4,600	ft ³	\$0.77	\$ 3,533
Grating	8,600	ft ²	\$0.18	\$ 1,548
Fencing	700	lf	\$1.94	\$ 1,358
Total Steel Demolition and Transport Cost - Burning Portion				\$ 819,512

Table 3
Summary of Concrete Volumes

CEMEX, Inc Lyons Mine
Permit No. M-1977-208
Bond Calculation

Structure	Quantity	Grinding Process		Total Cost
		Units	Cost	
Pavement demolition	600,000	s.f.	\$1.50	\$900,000
Pavement disposal	22,222	c.y.	\$10	\$222,222
Large foundations demolition	2,800	c.y.	\$30	\$84,000
Large foundations disposal	2,800	c.y.	\$10	\$28,000
Buildings demolition	1,712,001	c.f.	\$0.20	\$335,552
Silos demolition	8,499	c.y.	\$70	\$594,930
Silos disposal	8,499	c.y.	\$10	\$84,990
<u>Total Cost Grinding Process</u>				<u>\$2,249,694</u>

Structure	Quantity	Burning Process		Total Cost
		Units	Cost	
Pavement demolition	600,000	s.f.	\$1.50	\$900,000
Pavement disposal	22,222	c.y.	\$10	\$222,222
Large foundations demolition	3,768	c.y.	\$30	\$113,040
Large foundations disposal	3,768	c.y.	\$10	\$37,680
Buildings demolition	1,185,760	c.f.	\$0.20	\$232,409
Silos demolition	4,070	c.y.	\$70	\$284,900
Silos disposal	4,070	c.y.	\$10	\$40,700
<u>Total Cost Burning Process</u>				<u>\$1,830,951</u>

Table 4
Volumes of Concrete in Grinding Area

CEMEX, Inc Lyons Mine
Permit No. M-1977-208
Bond Calculation

Grinding Area Structures		Silos (yd ³)	Buildings (yd ³)	Foundations (yd ³)
Administration Building				
	Exterior walls		236	
	Interior Walls		256	
	Additional Floors			7
	Roof		233	
Mill Building				
	Exterior walls		733	
	Interior Walls		31	
	Additional Floors			114
	Roof		416	
	Beams			189
	Columns			320
	Mill Foundations			1,319
	Additional Foundations			13
Control Center				
	Exterior walls			
	Interior Walls			
	Additional Floors			160
	Roof		103	
Raw Material Silos		1,170		
Clinker Storage Building				
	Walls/Roof		2,629	
	Additional foundations			80
	Outside Clinker Pit			101
	Marginal Clinker			33
Pack House				
	Exterior walls		446	
	Additional Floors			442
	Roof		311	
	Columns			22
Finished Cement Silos		7,329		
Mobil Equipment Shed			190	
Electrical Room			38	
Wash House			8	
Pump House			37	
Total		8,499	5,667	2,800

Table 5
Volumes of Concrete in Burning Area

CEMEX, Inc Lyons Mine
Permit No. M-1977-208
Bond Calculation

Burning Area Structures		Silos (yd ³)	Buildings (yd ³)	Foundations (yd ³)
Burner Building	Exterior walls		509	
	Additional floors		199	
	Roof		223	
	Piers			3083
	columns			8
	Additional Foundations			35
Primary Crusher	Exterior walls		265	
	Divider wall			24
	Control bldg walls		7	
	Roof		416	
Secondary Crusher	Silos	342		
	Additional floors			60
Homogenizing Silos		1637		
Coal	Coal Silo	1115		
	Coal Unloading Building		159	
	Additional Foundations			19
	Cyclone Foundation			32
Stack		976		
Bag House	Columns			276
	Walls		422	
Raw Material Dryer	Dryer Bldg Walls		30	
	Foundations			155
Additional Foundations				76
Total		4,070	2,230	3,768

Table 6
Airspace of Buildings

CEMEX, Inc Lyons Mine
Permit No. M-1977-208
Bond Calculation

Building	length	width	height	volume of building (ft^3)		Burning Demolition	Grinding Demolition
Bag house	130	60	45	351,000	burning	351,000	
Pack house	150	72	25	270,000	grinding		270,000
Pack house	45	41	36	66,420	grinding		66,420
Crusher control building	10	10	10	1,000	burning	1,000	
Administration building	120	65	24	187,200	grinding		187,200
Administration building	105	46	14	67,620	grinding		67,620
Mill Building	168	97	53	863,688	grinding		863,688
Burner Building	120	91	53	578,760	burning	578,760	
Burner Building	48	91	25	109,200	burning	109,200	
Control Center	88	37	38	123,728	grinding		123,728
Control Center	37	34	20	25,160	grinding		25,160
Pump house	25	25	15	9,375	grinding		9,375
Wash house	10	10	8	800	grinding		800
Coal Unloading Building	71	36	30	76,680	burning	76,680	
Labor shed	96	40	18	69,120	burning	69,120	
Electrical Room	26	15	15	5,850	grinding		5,850
Miobil Equipment Shed	96	32	30	92,160	grinding		92,160
Total						1,185,760	1,712,001

Table 7
Onsite Disposal Area Construction

CEMEX, Inc Lyons Mine
Permit No. M-1977-208
Bond Calculation

<u>Item</u>	<u>Quantity</u>	<u>Units</u>	<u>Cost</u>	<u>Total Cost</u>
Excavation	1,000	yd ³	\$ 3.00	\$ 3,000
Low Permeability Liner Placement	50,000	ft ²	\$ 0.40	\$ 20,000
Cap Placement	6,000	yd ³	\$ 15.00	\$ 90,000
Total Cost for Disposal Site Construction				\$ 113,000

CEMEX, Inc Lyons Mine Permit No. M-1977-208
Bond Calculation

Appendix A
DMG Response Letter Dated March 6, 2003
DMG Response Letter Dated April 7, 2003

STATE OF COLORADO

DIVISION OF MINERALS AND GEOLOGY

Department of Natural Resources

1313 Sherman St., Room 215

Denver, Colorado 80203

Phone: (303) 866-3567

FAX: (303) 832-8106

Bill Owens
GovernorGreg E. Walcher
Executive DirectorRonald W. Catary
Division Director

March 6, 2003

Mr. Steve Mossberg
CEMEX
P.O. Box 529
Lyons, CO 80540RE: CEMEX, Inc., Lyons Mine, Permit No. M-1977-208
Reclamation Cost for Demolition of the Cement Plant

Dear Mr. Mossberg,

The Division received the initial bond calculation for the Lyons Mine, Permit No. M-1977-208 conducted by CEMEX's consultant Banks and Gesso. The Division has the following comments regarding the estimate that will need to be addressed by the operator and incorporated into the reclamation cost estimate;

1. Banks and Gesso states that equipment, steel structures, conveyors and offices on the site have salvage values greater than the cost of removal. Therefore, the cost of equipment removal or demolition was not included in the bond calculation. The Division concurs that mining equipment and machinery stored on site does not need to be included in the reclamation cost estimate (and any contents within the buildings). However, structures such as offices, steel structures (gas tanks, utility lines, etc.) and conveyors need to be included in the reclamation cost estimate. The Division does not accept the face value of salvage credit for structures unless it meets the standards defined by Rule 4.12 of the Construction Materials Rules and Regulations.

Rule 4.12 specifically outlines the requirements for salvage credit. In order to determine the appropriate salvage credit for each structure, CEMEX will be required to provide an itemized listing of the appraised salvage value of each structure, not including the salvage value of fixtures contained therein (4.12.2(1)). The appraisal should be by a qualified independent appraiser (4.9.1(2)(b)(i) and 4.12.2(3)). CEMEX will also need to annually verify the value of the structure (4.12.2(1)) and to provide an annual fee to fund the Division's cost to establish and review the salvage value component of the bond (4.1.2(9)). In addition CEMEX would have to report on encumbrances (4.12.2(2)) and would not be allowed to remove the structure from the permit area without prior consent of the Board (4.12.1(c) and 4.9.3(c)) and would be required to insure the structures (4.12.1(c) and 4.9.3(c)). A copy of the regulations regarding salvage credit is included for reference.

In addition, 100 percent of the salvage value could not be deducted from the amount of the required bond. Rather, some percentage of the total salvage credit could be deducted to allow for contractor profit and for administrative costs. Rule 4.9.1(1) states that the Board or Office may accept interests in real or personal property as Financial Warranties of not

more than 85 percent of the estimated value of such property. Hence, the estimated salvage value minus the costs for contractor profit and administrative costs, would be further reduced by a factor of .85 before being applied to the amount of required bond.

CEMEX will need to comply with the requirements of Rule 4.12 if salvage credit is used to off set costs within the reclamation cost estimate. Without full compliance, the cost to remove these structures will need to be included in the estimate.

2. Banks and Gesso presented two estimates for demolishing the plant area. Cost Estimate A assumes that all concrete is crushed to various sizes, and that the material is sold as aggregate for a total cost of \$3.9 million. Cost Estimate B includes a total cost of the demolition and reclamation assuming that all material is demolished, stockpiled and covered and the area is revegetated. Cost Estimate B totals \$4.6 million. CEMEX has agreed to submit \$3.9 million, which reflects the cost associated with crushing and selling the material as an aggregate product. Please be aware that the intent of the financial warranty is to insure that the State of Colorado has the necessary funds to complete reclamation in accordance with the approved mining and reclamation plan in the event the bond is forfeited. The Division is unable to accept an estimate that requires the State of Colorado to process material to be sold as aggregate, thus in turn acting as a mine operator. The Division's jurisdiction is to reclaim the site to the standards set forth in the permit, and ensure that the financial warranty can achieve this goal.

Cost Estimate B includes a cost for demolition and disposal of the material on site in the amount of \$4.6 million. The Division believes this scenario should be applied to reflect the required financial warranty for the Lyons site. However, in order to approve this plan, some modification to the mining and reclamation plan may be required. This is discussed in more detail below.

3. In order for mining structures and debris to be disposed of within the permit area, the operator will need to revise the mining and reclamation plan. Specifically, the operator will need to revise the permit to designate a disposal location for the structures and concrete to be buried on site. In addition, the operator will need to commit to a cover depth, and the source for the cover material (including a cover depth for all foundations to remain in place). Without such designation, the Division will need to assume that all material is hauled and disposed of in an approved landfill off site. The cost for such disposal in an off site landfill is much higher than the cost proposed by Banks and Gesso.

The Division recently approved such a technical revision for the Henderson Mine and Mill (Permit No. M-1977-342) in order to reduce the structural demolition cost. Within the submittal, the operator designated a location for the debris, committed to 3' of overburden material over the disposal site and included information to protect the surface and groundwater by lining the disposal site with an impermeable material. The technical revision filed by Henderson substantially reduced the cost for demolishing structures at the mine.

If CEMEX proposes to dispose of the material within the affected area of the permit, then a technical revision to the permit would be required. In addition, the components of the reclamation cost estimate prepared by Banks and Gesso should also be included in the technical revision.

4. Banks and Gesso applied a unit cost of \$0.25/cf to demolish all buildings and structures on site. The Division has a more detailed unit cost for structural demolition depending on the composition of the structure, height and disposal location (within 50 feet or hauled to a landfill). If CEMEX proposes to demolish and bury structures on site, the cost could range from \$0.115 to \$0.150, less than the \$0.25 proposed by Banks and Gesso. However, if the structures need to be hauled to an approved landfill off site, the cost may range between \$0.694 and \$0.837, considerably higher than the Banks and Gesso estimate. Attached is a table of demolition costs for your reference.
5. CEMEX provided a bond calculation map that is very helpful in determining the nature and location of structures at the cement plant. However, the map did not contain a scale necessary to determine the size of the structures. Please notify the Division as to the scale of the map, or provide a new map depicting a scale.
6. On Table 2 of Estimate B, Banks and Gesso itemized the quantity and unit cost of concrete demolition in the amount of 2,698 cy for the grinding process area and 3,768 cy for the burning process area. However, Table 4 and Table 5 outlines the "concrete in place" for each structure (16,828 cy in the grinding area and 10,068 cy for the burning area), which is significantly higher than the amounts listed in Table 2. Please explain why the "concrete in place" amount was not used as a factor in the reclamation cost estimate.
7. Please itemize all hazardous waste and chemicals stored on site that will need to be removed upon reclamation.
8. The reclamation cost estimate needs to include additional factors that would be incurred by the Division in the event of bond forfeiture. A cost for mobilization/demobilization to and from the job site should be included in the estimate. In addition, a number of factors make up the indirect costs and include liability insurance (1.55% of direct), performance bond (1.05% of direct), profit (10% of direct), engineering work and/or contract/bid preparation work, and reclamation management and/or administration (5% of direct and mandated by Rule 6.3.4(2)(c)). Please incorporate these costs into the reclamation cost estimate.

Please address the above noted questions/concerns by April 11, 2003. If you have any questions, please do not hesitate to contact me at 303-866-4943.

Sincerely,



Erica S. Crosby
Environmental Protection Specialist

Enclosure(s)

cc: Carl Mount; DMG

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Construction Materials Rule 4

117(3)(V)(A)

(1) Upon commencement of production, the Operator may establish an individual reclamation fund, to be held by an independent trustee for the Board, upon such terms and conditions as the Board may prescribe, which trust fund shall be funded by periodic cash payments representing such fraction of receipts as will, in the opinion of the Board, provide assurance that funds will be available for reclamation.

(2) Prior to issuance of a permit, the Operator will provide another form of Financial Warranty as described herein. As the reclamation fund increases in value, this form of Financial Warranty may be decreased in value so long as the sum of Financial Warranties is that amount specified by the Board or required by the Act.

(3) In approving the Individual Reclamation Fund as a Financial Warranty, the Board or Office shall:

- (a) approve the form of the initial Financial Warranty;
- (b) fix the fraction of receipts to be held in trust;
- (c) identify the trustee to hold said funds for the Board;
- (d) prescribe the terms and conditions applicable to the Operator or Warrantor's payment of funds into said trust; and
- (e) prescribe the terms and conditions governing the trustee's handling of said funds.

4.12 SPECIFIC REQUIREMENTS FOR SALVAGE CREDIT

117(3)(V)

4.12.1 Requirements for Salvage Credit

A Financial Warranty based on Salvage Credit must meet the following requirements:

117(3)(V)(C)

(a) Project-related fixtures and equipment (excluding Rolling Stock) owned or to be owned by the Operator/Permittee or Applicant within the permit area will have a Salvage Value at least equal to the amount of the Financial Warranty, or the appropriate portion thereof;

118(4)(b)

118(4)(c)

(b) Existing liens and encumbrances applicable to said fixtures and equipment, other than liens in favor of the United States or the State of Colorado, any other state, and any political subdivisions, will be subordinated to the lien described in Section 34-32.5-118(4)(b) and (c), C.R.S. and Paragraph 4.19(6).

117(3)(V)(E)

(c) Said fixtures and equipment will be maintained in good operating condition, be properly insured against theft, loss, fire and vandalism, and will not be removed from the permit area without the prior consent of the Board. In addition, the Warrantor shall ensure that insurance premiums are always paid two (2) years in advance on said fixtures and equipment.

4.12.2 Determination of Salvage Credit

(1) The Operator/Permittee or Financial Warrantor shall provide the Office with appraisals, information regarding invoice price, current value, cost of demolition and/or removal, and any other information as is necessary to establish the Salvage Value of the particular Project-related fixtures and/or equipment for which Salvage Credit is sought as all or part of the Financial Warranty for a Permit.

(2) The Operator/Permittee or Financial Warrantor shall provide the Office with a list of all encumbrances, and shall affirm that no other encumbrances exist to the best of the Operator's or Financial Warrantor's knowledge and belief.

(3) The Office may request the Operator/Permittee or Financial Warrantor to provide additional reasonable information to support the claimed Salvage Value and/or costs associated with any Project-related fixture or equipment for which Salvage Credit is sought.

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Construction Materials Rule 4

(4) Ten (10) days prior to any Board hearing regarding a Permit application for which Salvage Credit is offered as all or part of the Financial Warranty, the Office shall inform the Operator/Permittee or Financial Warrantor of its opinion as to the amount or estimate of the amount of the Salvage Value attributable to the Project-related fixtures and equipment for which Salvage Credit is sought.

~~(5) At the hearing before the Board, the Office shall recommend an amount for Salvage Credit value.~~

(6) The Board shall, after considering the Office's recommendation, testimony offered by the Operator, Warrantor, or any other person, and facts adduced at the hearing, fix the amount of the Salvage Credit for the Project-related fixtures and equipment, and attach conditions, as may be appropriate, to annually verify the value of the Salvage Credit.

4.13 REDUCTION OF WARRANTY AMOUNT

117(4)(a)
117(4)(c)

4.13.1 Operator's Request for Reduction

(1) An Operator may request that the Office reduce the amount of the Financial Warranty required.

(2) Such a request must:

(a) be made in writing, separate from other correspondence;

(b) include an estimate of the actual cost to reclaim the site based on what it would cost an independent contractor to complete reclamation, including unit costs for reclamation activities as appropriate to the operation to comply with the provisions of Section 3.1 and the Permit's Reclamation Plan.

(3) Such request shall be processed as described in 4.15, for Exploration operations, or Section 4.16, for all other operations.

DEMOLITION MENU - BUILDINGS and SIMILAR STRUCTURES**INSTRUCTIONS:**

Using the keyboard arrow keys, position the cellpointer over the desired item description (Items in bold red text).
Press the ENTER key to record your selection.

DATA SOURCE: MEANS Site Work & Landscape Cost Data Bk. 2003, Sec. 02220 (Demolition). Disposal costs per CPMCSOS.

NOTE: CF = Cubic Feet volume of standing structure. Total cost should include demolition PLUS disposal PLUS any dump charges.

Costs shown do NOT include costs for removal of concrete curb walls or sub-grade foundations/footings.

CY Rubble Per CF est. at 60% of standing structure vol., e.g. 10' x 10' x 10' bldg. = 1,000 CF/27 = 37.04 CY = .25 = 9.26 CY rubble.

NOTE: Use the KEYBOARD, not the MOUSE, to make a selection.

...table continues beyond bottom of screen

ITEM DESCRIPTION	UNIT OF MEASURE	CY Rubble PER UNIT	UNIT COST
HELP			
BACK			
DEMOLITION and DISPOSAL - SINGLE STORY NON-CONCRETE BLDGS. (floor slab/foundation not incl.)			
Bldg. (SN) demo./on-site disposal in existing pit or cut - Max. 50 ft. push	CF	NA	\$0.115
Bldg. (SN) demo./on-site disposal in existing pit or cut - Max. 200 ft. push	CF	NA	\$0.121
Bldg. (SN) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	CF	NA	\$0.125
Bldg. (SN) demo./on-site disposal in excavated pit - Max. 50 ft. push	CF	NA	\$0.128
Bldg. (SN) demo./on-site disposal in excavated pit - Max. 200 ft. push	CF	NA	\$0.135
Bldg. (SN) demo./on-site disposal in excavated pit - Max. 10,000 ft. haul	CF	NA	\$0.139
Bldg. (SN) demo./off-site disposal in approved landfill - Max. 5 mile haul	CF	NA	\$0.694
Bldg. (SN) demo./off-site disposal in approved landfill - Max. 15 mile haul	CF	NA	\$0.709
Bldg. (SN) demo./off-site disposal in approved landfill - Max. 30 mile haul	CF	NA	\$0.744
Bldg. (SN) demo./off-site disposal in approved landfill - Max. 60 mile haul	CF	NA	\$0.815
DEMOLITION and DISPOSAL - SINGLE STORY CONCRETE BLDGS. (floor slab/foundation not incl.)			
Bldg. (SC) demo./on-site disposal in existing pit or cut - Max. 50 ft. push	CF	NA	\$0.149
Bldg. (SC) demo./on-site disposal in existing pit or cut - Max. 200 ft. push	CF	NA	\$0.163
Bldg. (SC) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	CF	NA	\$0.160
Bldg. (SC) demo./on-site disposal in excavated pit - Max. 50 ft. push	CF	NA	\$0.163
Bldg. (SC) demo./on-site disposal in excavated pit - Max. 200 ft. push	CF	NA	\$0.169
Bldg. (SC) demo./on-site disposal in excavated pit - Max. 10,000 ft. haul	CF	NA	\$0.173
Bldg. (SC) demo./off-site disposal in approved landfill - Max. 5 mile haul	CF	NA	\$0.728
Bldg. (SC) demo./off-site disposal in approved landfill - Max. 15 mile haul	CF	NA	\$0.743
Bldg. (SC) demo./off-site disposal in approved landfill - Max. 30 mile haul	CF	NA	\$0.778
Bldg. (SC) demo./off-site disposal in approved landfill - Max. 60 mile haul	CF	NA	\$0.848
DEMOLITION and DISPOSAL - MULTI-STORY NON-CONCRETE BLDGS. (floor slab/foundation not incl.)			
Bldg. (MN) demo./on-site disposal in existing pit or cut - Max. 50 ft. push	CF	NA	\$0.139
Bldg. (MN) demo./on-site disposal in existing pit or cut - Max. 200 ft. push	CF	NA	\$0.145
Bldg. (MN) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	CF	NA	\$0.150
Bldg. (MN) demo./on-site disposal in excavated pit - Max. 50 ft. push	CF	NA	\$0.153
Bldg. (MN) demo./on-site disposal in excavated pit - Max. 200 ft. push	CF	NA	\$0.159
Bldg. (MN) demo./on-site disposal in excavated pit - Max. 10,000 ft. haul	CF	NA	\$0.163
Bldg. (MN) demo./off-site disposal in approved landfill - Max. 5 mile haul	CF	NA	\$0.718
Bldg. (MN) demo./off-site disposal in approved landfill - Max. 15 mile haul	CF	NA	\$0.733
Bldg. (MN) demo./off-site disposal in approved landfill - Max. 30 mile haul	CF	NA	\$0.768
Bldg. (MN) demo./off-site disposal in approved landfill - Max. 60 mile haul	CF	NA	\$0.837
DEMOLITION and DISPOSAL - MULTI-STORY CONCRETE BLDGS. (floor slab/found. not incl.)			
Bldg. (MC) demo./on-site disposal in existing pit or cut - Max. 50 ft. push	CF	NA	\$0.188
Bldg. (MC) demo./on-site disposal in existing pit or cut - Max. 200 ft. push	CF	NA	\$0.182
Bldg. (MC) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	CF	NA	\$0.196
Bldg. (MC) demo./on-site disposal in excavated pit - Max. 50 ft. push	CF	NA	\$0.199
Bldg. (MC) demo./on-site disposal in excavated pit - Max. 200 ft. push	CF	NA	\$0.206
Bldg. (MC) demo./on-site disposal in excavated pit - Max. 10,000 ft. haul	CF	NA	\$0.210
Bldg. (MC) demo./off-site disposal in approved landfill - Max. 5 mile haul	CF	NA	\$0.765
Bldg. (MC) demo./off-site disposal in approved landfill - Max. 15 mile haul	CF	NA	\$0.780
Bldg. (MC) demo./off-site disposal in approved landfill - Max. 30 mile haul	CF	NA	\$0.815
Bldg. (MC) demo./off-site disposal in approved landfill - Max. 60 mile haul	CF	NA	\$0.884
DEMOLITION ONLY - CONVENTIONAL (floor/foundation removal and disposal costs not included)			
Demo. only, small or single buildings (single story) - Steel structures	CF	0.00926	\$0.111
Demo. only, small or single buildings (single story) - Concrete structures	CF	0.00926	\$0.146
Demo. only, small or single buildings (single story) - Masonry structures	CF	0.00926	\$0.111
Demo. only, small or single buildings (single story) - Wood structures	CF	0.00926	\$0.111
Demo. only, large urban projects (multi-story bldgs.) - Steel structures	CF	0.00926	\$0.136
Demo. only, large urban projects (multi-story bldgs.) - Concrete structures	CF	0.00926	\$0.182
Demo. only, large urban projects (multi-story bldgs.) - Masonry structures	CF	0.00926	\$0.136
Demo. only, large urban projects (multi-story bldgs.) - Mixture of types	CF	0.00926	\$0.136
DEMOLITION ONLY - EXPLOSIVE/IMPLOSIVE (floor/foundation removal and disposal costs not included)			
Explosive demolition, large projects - Steel structures	CF	0.00926	\$0.190
Explosive demolition, large projects - Concrete structures	CF	0.00926	\$0.190
Explosive demolition, large projects - Masonry structures	CF	0.00926	\$0.190

STATE OF COLORADO

DIVISION OF MINERALS AND GEOLOGY

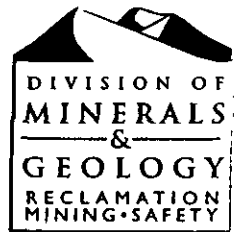
Department of Natural Resources

1313 Sherman St., Room 215

Denver, Colorado 80203

Phone: (303) 866-3567

FAX: (303) 832-8106



RECEIVED

APR 9 2003

BANKS & GESSO, LLC

Bill Owens
Governor

Greg E. Walcher
Executive Director

Ronald W. Cattany
Division Director

April 7, 2003

Steve Mossberg
CEMEX Inc.
P.O. Box 529
Lyons, CO 80540

**RE: *Reclamation Cost Estimate; Disposal of Concrete,
Lyons Quarry, Permit No. M-1977-208***

Dear Mr. Mossberg,

The Division recently spoke with Danelle Volpe of Banks and Gesso regarding the reclamation cost estimate for the Lyons Quarry, Permit No. M-1977-208. The conversation discussed reclamation costs and requirements for the disposal of concrete on site. The letter sent to you on April 2, 2003 addressed disposal for all structures at the CEMEX site (including concrete, steel, etc.). If CEMEX proposes to only dispose of concrete on site, then the Division would not require extensive testing of material or lining of the pit. The cost associated with the landfilling of concrete is enormous, especially for a site of this magnitude. If CEMEX only proposes to backfill concrete on site, the following criteria will need to be met:

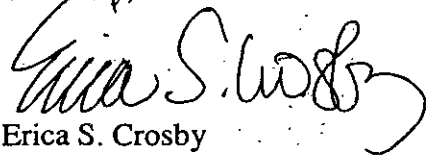
1. A technical revision to the permit must be submitted within 60-days of posting of the financial warranty.
2. A description of the disposal site(s) and their proximity and relationship to ground and surface water. A description of any ground and surface water protection measure to be installed such as landfill liners, covers, diversion ditches, or temporary storm water controls (*if needed*).
3. A description of the debris to be disposed on site, and a description of the types of material to be hauled to a landfill.
4. Provide an estimate of the volume of debris to be landfilled.
5. Provide a brief discussion of the hazardous materials that may be encountered during plant demolition and how they will be managed and disposed.
6. Provide a plan describing how debris will be placed and stabilized in disposal cells to prevent voids and settlement.

7. Locate the disposal sites on a map and provide maps, cross sections, and descriptions of the final configuration of disposal cells. The final graded surfaces of the debris must be free draining.
8. Provide a description of the cover soil that will cap the debris and a map and/or description of the source areas for the cover soil.

The Division hopes that this revised criteria list assists CEMEX in the submittal of the technical revision for the disposal of concrete-only material on site. In addition, the Division will grant an extension to the submittal of reclamation cost estimate to April 21, 2003.

If you have any questions, please feel free to call me at 303-866-4943.

Sincerely,



Erica S. Crosby
Environmental Protection Specialist

cc: Carl Mount, DMG (*via email*)
M. Danelle Volpe; Banks & Gesso LLC (*via fax*)

CEMEX, Inc Lyons Mine Permit No. M-1977-208
Bond Calculation

Appendix B
Concrete Calculation Back-up

ADMIN BLDGWALLS - PRESTRESSED CONCRETE PANELS ^{OR} CONCRETE BLOCKPERIMETER WALLS

BLOCK WALL 262 LF X 8" X 14' HIGH
 COLUMNS 22 - 8" X 1.5 FT X 14' HIGH
 251 LF X 4" X 24' HIGH
 T'S 63 - 8" X 1.6' X 24' HIGH

INTERIOR

742 LF X 8" X 14' HIGH
 =

ADD'L FLOOR

33' X 12' X 6" = 7.3

ROOF - 120' X 65' X 6" = 144
 46' X 105' X 6" 89

91 yd³
 11 yd³
 74 yd³
 60 yd³

236 yd³

REFERENCE

DWG

41-1-A

41-A-A

256 yd³241 yd³MILL BUILDING

EXTERIOR WALLS - PRESTRESSED CONCRETE

2 - 120' X 53' X 4" =
 91' X 53' X 4" =
 60 - 1.6 X 53' X 8" =
 30 - 1.6 X 53' X 8" =

LOW ROOF

157 yd³
 60 yd³
 126 yd³
 63 yd³

405 yd³REFERENCE
DRAWINGS

2 - 48' X 91' X 4"
 25' X 91' X 4"
 78' X 91' X 4"
 24 - 1.6 X 53' X 8"
 6 - 1.6 X 53' X 8"
 20 - 1.6 X 53' X 8"

HIGH ROOF

(T'S)
 (T'S)
 (T'S)

108 yd³
 28 yd³
 88 yd³
 50 yd³
 13 yd³
 42 yd³

03-01-CC
 03-05-CC
 03-04-CC

328 yd³INTERIOR WALLS

1.5' X 31 LF X 18 FT HIGH 31 yd³

REFERENCE DWG

03-01-CC

INTERIOR FLOOR - MOTOR ROOM

35' X 35' X 6" 03-5-CC 23 yd³ 03-1-CC

1521 yd³

MILL BLDG. CONTINUED

PRECAST BEAMS

1.3' x 2.3' x 93 B-3, B-4
 1 x 2.2 x 93 B-5, B-6, B7
 1 x 1.8 x 66 B-1, B2

REF. DWG
 03-2-CC

10 yd³
 8 yd³
 4 yd³ } 22 yd³

COLUMNS

14 - 2'8" x 2'8" x 80' HIGH
 14 - 1' x 1' x 50' HIGH

REF. DWG.
 03-1-CC

295 yd³
 26 yd³ } 320 yd³

RAW MILL FOUNDATION

46' x 4' x 20' BOTTOM SLAB 136 yd³
 18' x 14' x 26' SYMETRIC DRIVE 243 yd³
 9' x 9' x 26' 78 yd³
 18' x 5' x 16' STATIONARY DRIVE 53 yd³
 16' x 5' x 16' MOVEABLE BEARING 47 yd³

REF. DWGS.
 03-8-CC
 03-10-CC

557 yd³

FINISH MILL FOUNDATION

88' x 4' x 28' BOTTOM SLAB 365 yd³
 20' x 13' x 28' SYMETRIC DRIVE 270 yd³
 5' x 9' x 28' 47 yd³
 16' x 5' x 17' STATIONARY BEARING 50 yd³
 16' x 5' x 10' MOVEABLE BEARING 30 yd³

REF. DWGS.
 03-8-CC
 03-9-CC

726 yd³

MILL BLDG ROOF

97' x 168' x 6" 301 yd³
 24 - 2' x 8" x 97' (T's) 115 yd³

03-1-CC

416 yd³

ADDITIONAL FOUNDATIONS

3 - 10' x 3' x 4' - No dwg 13 yd³

13 yd³

BEAMS SUPPORTING SEPARATOR FLOOR

2 - 5' x 3' x 86' 96 yd³
 5 - 3' x 3' x 43' 72 yd³

03-3-CC

167 yd³

SLAB AROUND SEPARATOR

1' x 75' x 48 - (3 * 9²π + 7.5²π + 20' x 10')

03-3-CC 91 yd³

= 2229 yd³

CONTROL CENTER

WALLS - CONCRETE BLOCK

2 - 37' x 38' ^{HIGH} x 8"
 2 - 88' - 38' ^{HIGH} x 8"
 2 - 34' x 20' ^{high} x 8" Penthouse
 2 - 37' x 20' ^{high} x 8" "

REF DWG
47-1-CC

69 yd³
165 yd³
34 yd³ 305 yd³
37 yd³

ADD'L FLOORS

2 - 88' x 37' x 4"
 44 - 8" x 2' x 37' (T's)

80 yd³ 160 yd³
80 yd³

ROOF

88' x 37' x 4" (also 3rd floor below penthouse)
 22 - 8" x 2' x 37' T's
 25' x 37' x 4" penthouse
 6 - 8" x 2' x 37' T's

40 yd³
40 yd³ 103 yd³
11
11 yd³

568 yd³

BURNER BUILDINGPIERS

#1	10' x 29' x 22'
#2	13' x 23' x 29'
#3	40' x 25' x 29'
	19' x 25' x 35'
#4	13' x 23' x 35'
#5	13' x 24.5' x 38'

- Added later

236 yd³
321
1074
616
388
448

REF. DWG.

OS-1-CC

OS-3-CC

3083 yd³COLUMNS

11 - 2' x 2' x 63'

REF DWG.

OS-1-M

8 yd³ADD'L ENDS

4 -	3' x 4' x 12'
3 -	2' x 10' x 3'
	6 x 10 x 3

21 yd³
7
7

No drawing

35 yd³ROOF

69' x 4" x 128'
18 - 8" x 2' x 128'

T's

109 yd³
114 yd³

OS-12-CC

223 yd³WALLS

2 -	128 x 4" x 53'
2 -	69' x 4" x 53'
64 -	8" x 2' x 53'
32 -	8" x 2' x 53'

T's

T's

168 yd³
90 yd³
168 yd³
84 yd³

REF DWG

OS-12-CC

509 yd³INTERIOR FLOOR

1' x 47' x 114

OS-12-CC

199 yd³4056 yd³

PRIMARY CRUSHER

WALLS

WALLS - 2 - 43' x 2' x 30' high
REAR 20' x 2' x 30' high
PIERS 6 - 1.1' x 0.9' x 6' high
5' x 1.0' x 18'

WINGS

2 - $\frac{1}{2} (18 \times 29) \times 1.3'$

ROOF - CRUSHER

32' x 22' x 4" 9 yd³
8 - 8" x 2' x 22' T's 9

REF DWGS 191 yd³

02-2-CC 44

02-1-CC 1

02-3-CC 3

25

265 yd³

18 yd³

CONTROL BLDG

3 - 10' x 10' x 8" WALLS

200 yd³

DIVIDER WALL

23' x 2' x 14'

24 yd³

507 yd³

FLOORS/ LOTS - ASSUME ALL AREA IS PAVED 1FT THICK

SECONDARY CRUSHER

SEE "SILOS" FOR EXTERIOR CALC'S

ADD'L FLOORS

2 - $\left(\frac{32}{2}\right)^2 \pi \times 1' - \text{CRUSHER SILO}$

REFERENCE DWG
02-6-M

60yd³

RAW MATERIAL DRYER

DRYER BLDG

2 - 21' x 16' ^{high} x 8"
2 - 16' x 16' x 8"

17yd³
13

02-9-CC
02-10-CC

30yd³

DRYER FOUNDATION

5' x 15.5' x 13' 37
16.5' x 9' x 13' 72
7' x 9' x 2.5' 4
15.5' x 5' x 14' COLUMN 40

155yd³

244yd³

BOND CEMENT
CEMEX

7/10

SILOS

NAME	DIAM	WALL THK	HEIGHT	VOLUME
RAW MAT'L 1	27'	7"	73'	130 yd ³
REF 2-4-M	↓	↓	↓	↓
CLINKER 1	27'	7"	73'	130 yd ³
2				↓
3				
STACK (NO DWG)	44' OCTAGON	12"	195'	976
HOMOGEN. SILOS - HOMOGEN.	40'	1.5'	142'	954
REF KILN	31'	1.33'	142'	614
04-1,2,3,4-CL	20'	7"	52'	69
DUST				
COAL SILO	50'	0.9'	135'	707

ADD'L CONCRETE:

PLAN



$$\text{Volume of cylinder} = \pi r^2 h = \pi (25)^2 (23) = 45,160$$

$$\text{Vol. of cone} = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi (25)^2 (23) = 15,053$$

$$\text{Vol cylinder} - \text{volume of cone} = 115 \text{ yd}^3$$

NAME	DIAM	WALL THICK	HEIGHT	VOL
FINISHED CEMENT A1	20' OD	1.0'	150'	332
A2	↓	↓	↓	↓
A4	↓	↓	↓	↓
REF DWGS A5	↓	↓	↓	↓
08-6-CL/ A7	40'	1.67'		1115
08-5-CL/ A9	20'	1.0'		332
08-4-CL A10	↓	↓	↓	↓
A12	↓	↓	↓	↓
A13	↓	↓	↓	↓
B1	40'	1.67'		1115
B2	20'	1.0		332
B3	↓	↓	↓	↓
B5	↓	↓	↓	↓
B6	↓	↓	↓	↓
B7	40'	1.67		1115
SECONDARY CRUSHER DUST	26' (ID)	7"	85	750
02-5-M CRUSHER	32	8"	55	934
02-8-CL SURGE	20	7"	44	58
				12,125 yd ³

BAGHOUSE

COLUMNS

30 - 1.75' x 1.75' x 21'
40 - 1.75' x 1.75' x 45'

72 yd³
204 yd³

05-17-CC

276 yd³

WALLS

2 - 60' x 8" x 45'
2 - 130' x 8" x 45'

133 yd³
289

422 yd³

PACKHOUSE

WALLS

2 - 150' x 4" x 25'
75 - 8" x 2' x 25'
2 - 72' x 4" x 25'
36 - 8" x 2' x 25'
2 - 41' x 4" x 45'
20 - 8" x 2' x 45'
2 - 36' x 4" x 45'
9 - 8" x 2' x 45'

7's
7's
7's
7's
7's
7's
7's

93 yd³
93 yd³
44 yd³
44 yd³
46 yd³
46 yd³
40 yd³
40 yd³

REF DWGS

09-1-CC

09-3-CC

446 yd³

INTERIOR FLOORS

3 - 41' x 36' x 1'
1 - 100' x 75' x 1'

164 yd³
278 yd³

442 yd³

ROOF

150' x 84' x 4"
21 - 8" x 2' x 150'

156 yd³
156 yd³

311 yd³

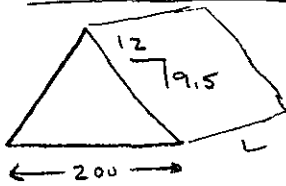
COLUMNS

2 - 2' x 1' x 25'
2 - 2' x 1.5' x 70'
1 - 1' x 1' x 70'

4
16
3

22 yd³

1919 yd³

CLINKER STORAGE BLDG

$$L = 388'$$

$$\begin{aligned} & \frac{1}{2}(2) 100 (79.2) \times 4'' - \text{CLOSED END} & 98 \text{ yd}^3 \\ & 25 - 8'' \times 2' \times 79' & 7's & 98 \text{ yd}^3 \\ & 2 - 127 \times 388 \times 4'' & \text{ROOFS} & 1217 \text{ yd}^3 \\ & 194 - 8'' \times 2' \times 127 & 7's & 1217 \text{ yd}^3 \\ & & & \underline{2629 \text{ yd}^3} \end{aligned}$$

CONVEYOR - ABOVE GROUND

$$\begin{aligned} & 50 \text{ yd}^3 - \text{TUNNEL} \\ & 30 \text{ yd}^3 - \text{HOPPER} \end{aligned} \quad \text{NO DWG}$$

$$80 \text{ yd}^3$$

COAL UNLOADING BLDG

$$\begin{aligned} & 2 - 71' \times 30' \times 4'' & 53 \text{ yd}^3 \\ & 36 - 2' \times 8'' \times 30' & 7's & 53 \text{ yd}^3 \\ & 2 - 36' \times 30' \times 4'' & & 27 \text{ yd}^3 \\ & 9 - 2' \times 8'' \times 30' & 7's & 27 \text{ yd}^3 \end{aligned} \quad \text{NO DWG} \quad 159 \text{ yd}^3$$

LABOR SHED

$$\begin{aligned} & 2 - 96' \times 18' \times 4'' & 43 \text{ yd}^3 \\ & 48 - 2' \times 8'' \times 18' & 43 \text{ yd}^3 \\ & 2 - 40' \times 18' \times 4'' & 18 \text{ yd}^3 \\ & 20 - 2' \times 8'' \times 18' & 18 \text{ yd}^3 \end{aligned} \quad 122 \text{ yd}^3$$

MOBILE EQUIPMENT SHED

$$\begin{aligned} & 2 - 32' \times 30' \times 4'' & 24 \text{ yd}^3 \\ & 16 - 8' \times 2' \times 30' & 24 \text{ yd}^3 \\ & 2 - 96' \times 30' \times 4'' & 71 \text{ yd}^3 \\ & 48 - 2' \times 8'' \times 30' & 71 \text{ yd}^3 \end{aligned} \quad 190 \text{ yd}^3$$

ELECTRICAL ROOM

$$\begin{aligned} & 2 - 22' \times 15' \times 4'' & 9 \text{ yd}^3 \\ & 12 - 8'' \times 2' \times 15' & 9 \text{ yd}^3 \\ & 2 - 26' \times 15' \times 4'' & 10 \text{ yd}^3 \\ & 14 - 8' \times 2' \times 15' & 10 \text{ yd}^3 \end{aligned} \quad 38 \text{ yd}^3$$

$$\underline{3217 \text{ yd}^3}$$

PUMP HOUSE

4 - 25 x 15 x 8"

NO DWGS

37 yd³ADDL FNDNSCOAL CYCLONE

18' x 24' x 2'

32 yd³DRYER FOUNDATION

3.5 x 9' x 16'

19 yd³PIERS

6 - 2' x 1.5' x 7'

4' x 10' x 10'

4 - 5' x 2.5' x 2.5'

2 - 3' x 2.5' x 2.5'

13 - 2.5 x 2.5 x 12'

10' x 5' x 4'

2 - 2.5 x 10' x 4'

DUST COLLECTOR
FOUNDATIONS
ALONG EAST SIDE

FAN

5 yd³15 yd³5 yd³1 yd³36 yd³7 yd³7 yd³76 yd³OUTSIDE CLINKER PT

2 - 21' x 15' x 15'

27' x 15' x 15'

27' x 15' x 15' - (25 x 14 x 14)

35 yd³22 yd³44 yd³101 yd³MARGINAL CLINKER

3 - 20' x 15' x 1

33 yd³COAL SILO

10' x 10' x 5'

19 yd³WASH HOUSE

4 - 10' x 8' x 8"

8 yd³325 yd³

CEMEX, Inc Lyons Mine Permit No. M-1977-208
Bond Calculation

Appendix C
Steel Calculation Back-up

Tanks/Hoppers

$$\text{surf. area} = \pi r h$$

Mill Bldg -

Silo Hoppers - assume h = ϕ
- assume each silo has steel hopper of 1/8" metal

G	15	-	27' ϕ	-	$\pi (27/2)^2 * 27 = 15,459 \text{ ft}^3$	$\times 15 = 231,885 \text{ ft}^3$
G	12	-	20' ϕ	-		75,398
G	4	-	40' ϕ	-		201,061
B	1	-	30' ϕ	-		21,205
B	1	-	26' ϕ	-		13,804
B	1	-	32' ϕ	-		25,736
B	1	-	21' ϕ	-		8724
B	1	-	20' ϕ	-		6283

Baghouse hoppers

$$B \quad 13 - 10' \phi \quad = 2042 \text{ ft}^3$$

$$B \quad 6 \text{ rows} - \quad =$$

$$70' \text{ long, } 6' \text{ high "V"}$$

$$6 \times 70 \times \frac{1}{2} (6)(6) = 7560 \text{ ft}^3$$

$$\text{Total Grinding} = 508344 \text{ ft}^3$$

$$\text{Total Burning} = 85,357 \text{ ft}^3$$

Utility Lines

$$\text{Total Distance} \quad 0.5 \text{ mi} - 2640 \text{ LF}$$

Conveyors

Coal conveyor	200 LF	B -	Grinding = 800 LF
Clinker conveyor	300 LF	G	
Crusher conveyors	500 LF	B -	
Kiln conveyor	800 LF	B -	Burning = 2700 LF
Kiln conveyor	800 LF	B -	
Crusher conveyor	400 LF	B -	
Conveyor to raw silos	500 LF	G	

Rubble produced is 0.555 yd³/LF

Stairs 1 LF of stairs = 3 cu. ft of air space

G	Clinker	500 LF	= 1500 ft ³	
B	Coal	100 LF	300	
G	Mill Bldg	300 LF	900	Total Burning =
G	Silos	400 LF	1200	4575 ft ³
B	Burning / Cooling	200 LF	600	⇒ 4600 ft ³
G	Cement Storage	300 LF	900	
B -	Foundation #3 burner =	100 LF	300	Total Grinding =
G -	Raw Material =	50 LF	150	18,270 ft ³
B	Dryer =	60 LF =	180	⇒ 18,300 ft ³
G	Dust collectors =	250 LF	750	
	Silos Raw Matl			
G	12 - 2x 27'φ =	7040 LF	6120	
	Packhouse			
G	12 - 2x 20'φ =	1500 LF	4500	
G	3 - 2x 40'φ =	750 LF =	2250	
	Homog.			
B	2x 40'φ =	250 LF	750	
B	2x 30'φ =	200 LF	600	
B	2x 21'φ =	130 LF	390	
	Crusher 2nd			
B	2x 26'φ	160 LF	480	
B	2x 32'φ	200 LF	600	
B	2x 20'φ	125 LF	375	

Metal Bldgs / Structures

B	Preheater Tower	55' x 40' x 60' =	132,000	ft ³
B/G	Electrical Room	25' x 20' x 20' =	10,000	ft ³
B	Kiln Bag house	55' x 60' x 70' =	231,000	ft ³
B	Alkali Bag house	45' x 50' x 70' =	157,500	ft ³
G	Add'l Dust Suppressor Structures	70 x 30 x 60' =	126,000	ft ³
B	Burner Foundation #3	40' x 40' x 60' =	96,000	ft ³

Total Burning = 747,500 ft³

Total Grinding = 131,000 ft³

Fencing

G	Silos	300 LF x 8'	
G	Raw Mat'l	100 LF x 8'	
1/2G/1/2B	Walkways	600 LF x 8'	

Total Grinding = 700 LF Total Burning = 300 LF

Grating

3/4G/1/4B	Silos exterior	3050 ft ²
3/4G/1/4B	Silos Interior	19,067 ft ²
1/2G/1/2B	Walkways	1000 LF x 6' = 6000 ft ²

Total Grinding = 19,587 ft²
⇒ 19,600 ft²

Total Burning = 8529 ft²
⇒ 8600 ft²

Appendix D
Itemization of Hazardous Waste and Chemicals

List of Chemicals Stored on Site as of March 2003

Ammonium Nitrate
Hydrochloric acid
Acetic acid
Nitric acid
Ammonium Hydroxide
Sodium Hydroxide
Methanol.